

The Effect of Wildfire Smoke on Academic Achievement

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What the paper does

- ▶ Effect of wildfire smoke on reading and numeracy, B.C. grades 4 and 7
- ▶ Population of B.C. public and independent school students, 2010/11-2021/22:
 - ▶ Foundation Skills Assessment (FSA) scores
 - ▶ Parents' income (T1FF tax files)
 - ▶ Heavy-smoke days (NOAA HMS) and smoke $PM_{2.5}$ (CanOSSEM), by six-digit postal code
- ▶ Identification: year-to-year variation in smoke within postal codes
 - ▶ Falsification: future smoke; excluding students near fire perimeters
- ▶ Ten heavy-smoke days in the 12 months before the test lower scores by 0.03-0.05 SD
 - ▶ Concentrated among boys; summer timing read as a physiological channel

Wildfire smoke, policy and the literature

Policy

- ▶ Since 2016, smoke eroded 23% of prior $PM_{2.5}$ gains in two-thirds of U.S. states (Burke et al. 2023)
- ▶ 2023 Canadian fires: $+3.8 \mu g/m^3$ national annual-mean $PM_{2.5}$; 64,300 chronic deaths (Zhang et al. 2025)
- ▶ Documented costs of smoke: earnings, emergency visits, suicide (Borgschulte et al. 2024; Heft-Neal et al. 2023; Molitor et al. 2023)

Literature

- ▶ Air pollution and learning (a fairly large literature):
 - ▶ Exam day (Ebenstein et al. 2016; Graff Zivin et al. 2020)
 - ▶ School year (Heissel et al. 2022; Gilraine and Zheng 2024)
 - ▶ Prenatal (Sanders 2012; Bharadwaj et al. 2017)
- ▶ Wildfire smoke and achievement: Wen and Burke (2022); Behrer et al. (2026); Arrizaga et al. (2026)
- ▶ Open question: lasting human capital or transient performance?

Magnitude and positioning

- ▶ Per $\mu\text{g}/\text{m}^3$ -day, of similar magnitude to Wen and Burke (2022) and Gilraine and Zheng (2024)
 - ▶ One B.C. heavy-smoke day = about 65 $\mu\text{g}/\text{m}^3$ -days of smoke $\text{PM}_{2.5}$
 - ▶ Ten heavy days = about 19 average U.S. district smoke-years
- ▶ No sign of a smoke-specific toxicity premium for test scores
- ▶ Individual-level data now also in Behrer et al. (2026): about 10 million U.S. PSAT takers
- ▶ Build the contribution on what studying BC adds to the literature:
 - ▶ Extreme doses, above Behrer et al.'s moderate days (8-12 $\mu\text{g}/\text{m}^3$)
 - ▶ The 2017 test-timing reform
 - ▶ A monetized Canadian human-capital cost

The summer puzzle

- ▶ Smoke arrives in July-August, when school is out, yet tests are administered in fall/winter
- ▶ How can summer exposure enter grades?
- ▶ School-year exposure has a clear channel:
 - ▶ Smoke $PM_{2.5}$: the effect comes from school-year days (Behrer et al. 2026)
 - ▶ Closures and absences on smoke days (Graff Zivin and Lee 2026)
 - ▶ Heat: school-year heat lowers learning; summer heat does little (Park et al. 2020)
- ▶ Summer smoke lowers next-year GPA in Chile, with no identifiable attendance effect (Arrizaga et al. 2026)
- ▶ The paper's own timing: effect 2-4 months after the season, none at 5-7 months

The summer puzzle: evidence from U.S. data

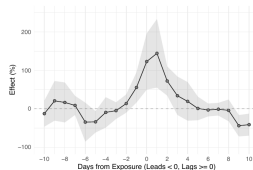
Behrer et al. (2026): U.S. PSAT retakes

Table 7: Instrumental Variables - Impact of School Year and Summer Pollution

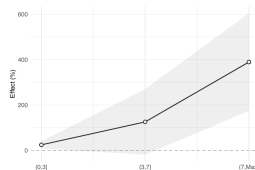
	(1)	(2)	(3)
(A) Overall pollution			
School year pollution days	-0.097 (0.068)		-0.186** (0.078)
Summer pollution days		-0.039 (0.193)	0.329 (0.234)
(B) By pollution intensity			
School year moderate pollution days	-0.347*** (0.105)		-0.702** (0.325)
School year heavy pollution days	-0.032 (0.067)		-0.152 (0.137)
Summer moderate pollution days		-2.228*** (0.411)	0.587 (1.009)
Summer heavy pollution days		-0.603*** (0.218)	0.551 (0.486)
N	16,742,701	16,742,701	16,742,701

Panel B: summer days statistically significant alone (col. 2); statistically insignificant once school-year days enter (col. 3). Source: Behrer et al. (2026), Table 7

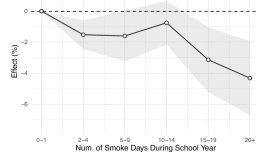
Graff Zivin and Lee (2026): closures (CA), attendance (TX)



(a) Smoke and School Closure over Time



(b) Smoke and School Closure by PM Intensity



(c) Smoke and Student Attendance

Closure probability rises from 3.6% to about 18% on high-smoke days; attendance 4.5% lower with 20+ smoke days. Source: Graff Zivin and Lee (2026), Figure 3

Check summer vs. school-year exposure?

- ▶ Does ANY smoke day overlap with the school year?
- ▶ Split exposure into September and July-August days, entered jointly
- ▶ Monthly distributed lag by months before the test, pooling both testing regimes

Candidate mechanisms for summer exposure

- ▶ Physiological: respiratory illness, inflammation
 - ▶ Larger for students with a chronic-health designation?
- ▶ Psychological: child mental health, stronger for boys (Zhang et al. 2026)
 - ▶ Mental-health designations: precisely estimated zeros, though they respond to parental earnings losses (Jones et al. 2024)
- ▶ Disrupted summer: children indoors, lost summer camp
 - ▶ Not sure how much summer camp attendance correlates with school grades
- ▶ Household income: smoke lowers parents' earnings (Borgschulte et al. 2024)
 - ▶ Smoke effect on parental earnings through T1 filing?